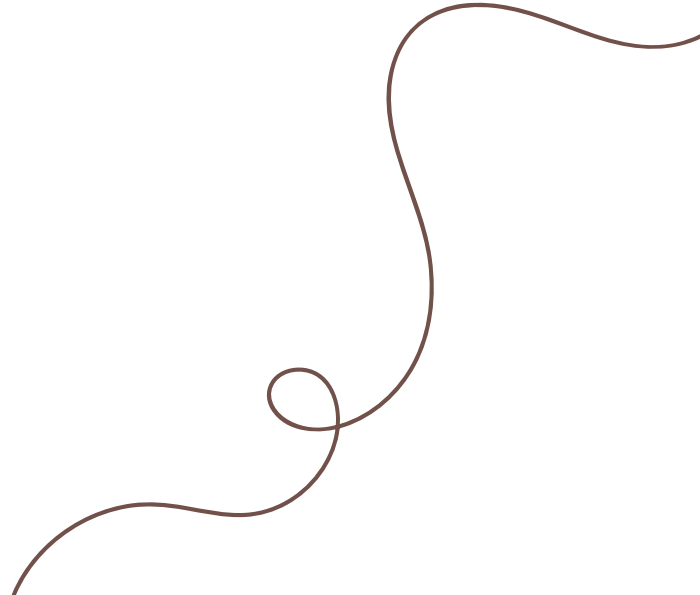




Chocolate Sales Analysis

Sales & Revenue Insights (2023–2024)

*presented by
Aungsangui*



Title: Dataset Overview

Source: Kaggle Chocolate
Sales Dataset

Time Period: 2023 – 2024

Database used: Oracle
Database

Analysis tool: SQL



○ Tables Used:

1. Customers

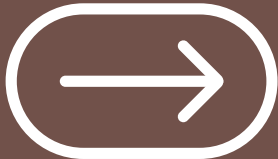
2. Stores

3. Sales

4. Calendar



Show all the tables-1

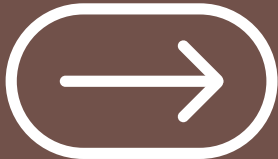


Select * from customer
Fetch first 15 rows only;



	CUSTOMER_ID	AGE	GENDER	LOYALTY_MEMBER	JOIN_DATE
1	C000001	40	Male	1	21-05-25
2	C000002	47	Male	0	26-12-21
3	C000003	58	Female	1	13-09-22
4	C000004	25	Female	0	27-02-25
5	C000005	43	Male	0	31-08-23
6	C000006	32	Male	0	22-05-22
7	C000007	51	Male	1	28-07-24
8	C000008	56	Female	0	12-08-24
9	C000009	18	Male	0	23-06-25
10	C000010	40	Male	0	07-06-23
11	C000011	44	Male	0	22-12-25
12	C000012	53	Male	1	24-01-24
13	C000013	68	Male	1	23-03-22
14	C000014	32	Female	1	16-08-24
15	C000015	61	Female	1	18-02-26

Show all the tables-2

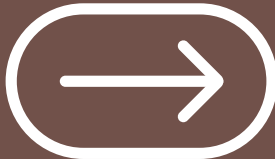


Select * from calender
Fetch first 10 rows only;



	↕ C_DATE	↕ YEAR	↕ MONTH	↕ DAY	↕ WEEK	↕ DAY_OF_WEEK	
1	01-01-23	2023	1	1	52	6	
2	02-01-23	2023	1	2	1	0	
3	03-01-23	2023	1	3	1	1	
4	04-01-23	2023	1	4	1	2	
5	05-01-23	2023	1	5	1	3	
6	06-01-23	2023	1	6	1	4	
7	07-01-23	2023	1	7	1	5	
8	08-01-23	2023	1	8	1	6	
9	09-01-23	2023	1	9	2	0	
10	10-01-23	2023	1	10	2	1	

Show all the tables-3

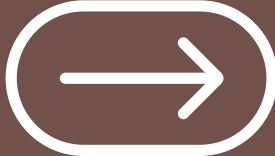


Select * from sales
Fetch first 15 rows only;

	ORDER_ID	ORDER_DATE	PRODUCT_ID	STORE_ID	CUSTOMER_ID	QUANTITY	UNIT_PRICE	DISCOUNT	REVENUE	COST	PROFIT
1	ORD000000001	07-01-23	P0080	S093	C040749	5	14.43	0.15	61.33	42.77	18.56
2	ORD000000002	22-10-23	P0173	S065	C020161	3	12.01	0	36.03	19.06	16.97
3	ORD000000003	07-05-23	P0115	S078	C048069	2	10.02	0	20.04	10.29	9.75
4	ORD000000004	23-06-24	P0186	S088	C047901	2	14.66	0.1	26.39	16.35	10.04
5	ORD000000005	24-09-24	P0197	S054	C033950	1	12.34	0	12.34	7.94	4.4
6	ORD000000006	29-03-24	P0160	S089	C008918	4	13.52	0	54.08	36.59	17.49
7	ORD000000007	26-02-23	P0062	S024	C002897	1	11.97	0.1	10.77	7.16	3.61
8	ORD000000008	03-11-23	P0111	S085	C038072	5	4.62	0	23.1	16.15	6.95
9	ORD000000009	11-10-24	P0135	S029	C003786	4	7.88	0	31.52	19.9	11.62
10	ORD000000010	17-12-23	P0069	S056	C043148	3	8.88	0	26.64	18.19	8.45
11	ORD000000011	01-01-23	P0181	S070	C016710	5	12.62	0	63.1	43.74	19.36
12	ORD000000012	18-03-23	P0137	S028	C040620	4	4.1	0	16.4	9.9	6.5
13	ORD000000013	06-11-24	P0003	S021	C048449	3	10.9	0	32.7	17.29	15.41
14	ORD000000014	08-11-24	P0026	S005	C009995	2	4.43	0.1	7.97	5.5	2.48
15	ORD000000015	08-01-24	P0042	S100	C011763	2	5.48	0	10.96	6.83	4.13



Show all the tables-4

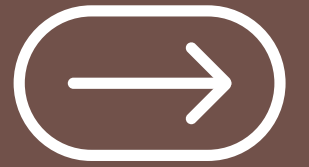


Select * from stores
Fetch first 15 rows only;



	⚡ STORE_ID	⚡ STORE_NAME	⚡ CITY	⚡ COUNTRY	⚡ STORE_TYPE
1	S001	Chocolate Store 1	New York	Canada	Retail
2	S002	Chocolate Store 2	Melbourne	Canada	Mall
3	S003	Chocolate Store 3	Berlin	France	Mall
4	S004	Chocolate Store 4	Paris	UK	Airport
5	S005	Chocolate Store 5	Sydney	USA	Online
6	S006	Chocolate Store 6	Toronto	Canada	Online
7	S007	Chocolate Store 7	Sydney	France	Mall
8	S008	Chocolate Store 8	Paris	UK	Mall
9	S009	Chocolate Store 9	Paris	France	Online
10	S010	Chocolate Store 10	Toronto	UK	Retail
11	S011	Chocolate Store 11	New York	USA	Retail
12	S012	Chocolate Store 12	London	France	Mall
13	S013	Chocolate Store 13	Melbourne	France	Retail
14	S014	Chocolate Store 14	London	Australia	Airport
15	S015	Chocolate Store 15	Paris	Canada	Airport

Average discount given

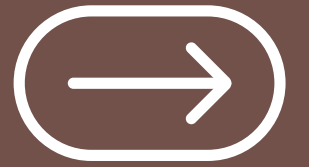


Select `round(avg(discount),2)` as
Average_discount from sales;

	AVERAGE_DISCOUNT
1	0.06



Total sales revenue

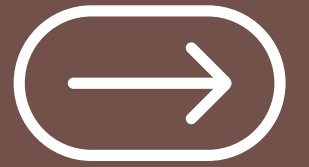


```
select round(sum(revenue), 2) as "Total  
Sales"  
from sales;
```

	Total Sales
1	25486128.86

o o o o

Total quantity of chocolate sold

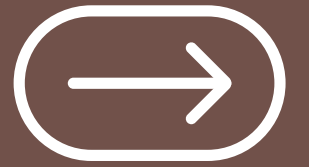


```
select sum(quantity) as "Total Quantity"  
from sales;
```

	Total Quantity
1	2999589

o o o o

Number of unique customers

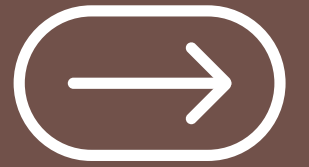


```
select count(distinct customer_id) as  
unique_customers  
from sales;
```

	UNIQUE_CUSTOMERS
1	50000

○ ○ ○ ○

Total no. of customers and total sales profit



select count (customer_id) as Customer_id
from sales;

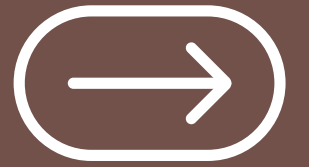
	CUSTOMER_ID
1	1000000

select sum(profit) as Total_profit from
sales;

	TOTAL_PROFIT
1	10194564.63



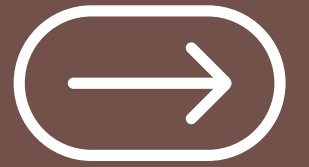
Average order value



```
select  
round(avg(revenue),2) as average_value  
from sales;
```

	AVERAGE_VALUE
1	25.49

Top 10 customers by sales

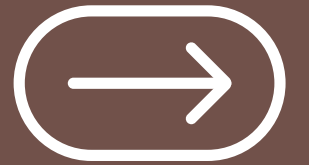


```
select
  customer_id,
  store_id,
  sum(revenue) as total_profit
from
  sales
group by
  customer_id,
  store_id
order by
  total_profit desc
fetch first 10 rows only;
```



	⌘ CUSTOMER_ID	⌘ STORE_ID	⌘ TOTAL_PROFIT	
1	C017975	S023	206.66	
2	C026473	S058	191.11	
3	C006970	S043	191.06	
4	C032377	S092	186.21	
5	C008775	S088	183.96	
6	C014567	S077	181.9	
7	C047117	S032	180.42	
8	C018482	S084	179.75	
9	C018789	S083	178.81	
10	C001911	S002	178.69	

Total sales by country

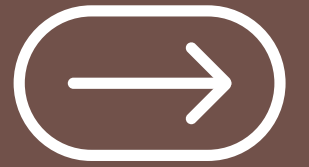


```
select
  c.country,
  sum(s.revenue) as
total_sales_by_country
from
  sales s
join stores c
on s.store_id = c.store_id
group by c.country
order by
total_sales_by_country desc;
```

○ ○ ○ ○

	⌘ COUNTRY	⌘ TOTAL_SALES_BY_COUNTRY	
1	Canada	5085319.05	
2	UK	4823617.98	
3	USA	4341494.26	
4	France	4339730.9	
5	Australia	3834803.4	
6	Germany	3061163.27	

Monthly sales trend

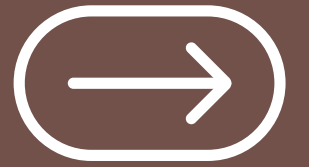


```
Select
  m.month,
  sum(s.revenue) as monthly_sales
from
  calender m
  join sales s on m.c_date = s.order_date
group by
  m.month
order by
  m.month ASC;
```



	MONTH	MONTHLY_SALES
1	1	2179770.2
2	2	1978991.17
3	3	2169216.91
4	4	2081878.46
5	5	2162061.97
6	6	2084259.26
7	7	2169146.47
8	8	2169043.54
9	9	2108349.78
10	10	2146961.85
11	11	2078753.22
12	12	2157696.03

Yearly sales

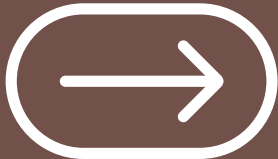


```
select
  y.year,
  sum(r.revenue) as yearly_sales
from
  calender y
  join sales r on y.c_date = r.order_date
group by
  y.year;
```

	YEAR	YEARLY_SALES	
1	2023	12735106.27	
2	2024	12751022.59	



Store with highest sales

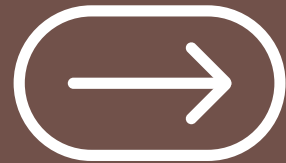


```
select
  s.store_name,
  sum(o.revenue) as highest_sales
from
  stores s
  join sales o on s.store_id = o.store_id
group by
  s.store_name
order by
  highest_sales desc
fetch first 5 rows only;
```



	STORE_NAME	HIGHEST_SALES
1	Chocolate Store 74	261393.77
2	Chocolate Store 33	260672.37
3	Chocolate Store 50	259526.62
4	Chocolate Store 85	259512.15
5	Chocolate Store 98	259055.13

Monthly sales per store

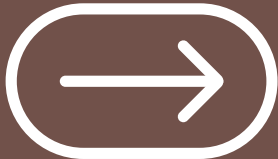


```
select m.month, sum(r.revenue) as  
monthly_sales, r.store_id, e.store_name  
from calender m  
join sales r  
on m.c_date = r.order_date  
join stores e  
on e.store_id = r.store_id  
group by m.month, r.store_id,  
e.store_name  
order by Monthly_sales desc  
fetch first 12 rows only;
```



	MONTH	MONTHLY_SALES	STORE_ID	STORE_NAME
1	8	24664.63	S006	Chocolate Store 6
2	3	24197.77	S009	Chocolate Store 9
3	8	23883.3	S030	Chocolate Store 30
4	7	23841.82	S040	Chocolate Store 40
5	1	23823.65	S033	Chocolate Store 33
6	10	23707.59	S021	Chocolate Store 21
7	1	23648.36	S051	Chocolate Store 51
8	7	23647.22	S065	Chocolate Store 65
9	10	23586.53	S018	Chocolate Store 18
10	3	23572.86	S071	Chocolate Store 71
11	9	23560.73	S096	Chocolate Store 96
12	1	23516.98	S060	Chocolate Store 60

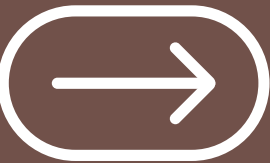
Top 5 products by revenue



```
select product_id, sum(revenue) as  
revenue_by_product  
from sales  
group by product_id  
order by Revenue_by_product desc  
fetch first 5 rows only;
```

	PRODUCT_ID	REVENUE_BY_PRODUCT
1	P0091	131813.61
2	P0107	131799.85
3	P0010	130258.58
4	P0008	130153.07
5	P0133	129789.85

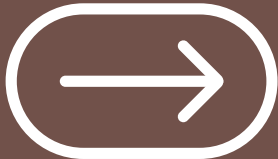
Rank stores by sales



```
select store_id, sum(revenue) as  
Total_Sales,  
Dense_rank() over (order by sum(revenue)  
desc ) as sales_rank from sales  
group by store_id  
fetch first 10 rows only;
```

	STORE_ID	TOTAL_SALES	SALES_RANK
1	S074	261393.77	1
2	S033	260672.37	2
3	S050	259526.62	3
4	S085	259512.15	4
5	S098	259055.13	5
6	S027	259006.23	6
7	S022	258914.68	7
8	S030	258231.17	8
9	S013	258185.98	9
10	S090	258135.85	10

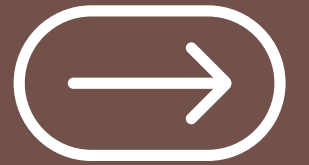
Top 3 customers per store



```
select * from
(Select customer_id, store_id,
sum(revenue) as total_sales,
Dense_rank() over (partition by store_id
order by sum(revenue) ) as Cust_rank from
Sales
group by customer_id, store_id)
where Cust_rank <=3
fetch first 11 rows only;
```

	CUSTOMER_ID	STORE_ID	TOTAL_SALES	CUST_RANK
1	C025297	S001	2.42	1
2	C048729	S001	2.42	1
3	C031535	S001	2.42	1
4	C025699	S001	2.43	2
5	C046990	S001	2.47	3
6	C016564	S002	2.41	1
7	C034239	S002	2.47	2
8	C035217	S002	2.53	3
9	C046593	S003	2.42	1
10	C036861	S003	2.43	2
11	C036385	S003	2.47	3

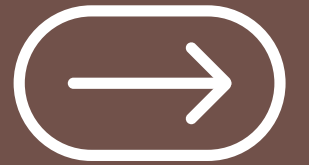
Highest sales by store type



```
Select s.store_type, sum(r.revenue) as  
Total_sales from sales r  
join stores s on r.store_id = s.store_id  
group by s.store_type order by  
sum(r.revenue) desc;
```

	STORE_TYPE	TOTAL_SALES
1	Airport	7613875.92
2	Mall	6628228.67
3	Online	6386035.76
4	Retail	4857988.51

Running total sales by month

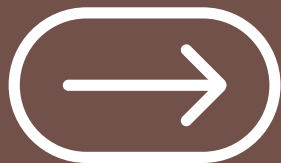


```
select c.month, sum(s.revenue) as  
total_sales,  
sum(sum(s.revenue) )over (order by  
c.month ) as Running_total  
from calender c  
join sales s on s.order_date = c.c_date  
group by c.month ;
```



	MONTH	TOTAL_SALES	RUNNING_TOTAL	
1	1	2179770.2	2179770.2	
2	2	1978991.17	4158761.37	
3	3	2169216.91	6327978.28	
4	4	2081878.46	8409856.74	
5	5	2162061.97	10571918.71	
6	6	2084259.26	12656177.97	
7	7	2169146.47	14825324.44	
8	8	2169043.54	16994367.98	
9	9	2108349.78	19102717.76	
10	10	2146961.85	21249679.61	
11	11	2078753.22	23328432.83	
12	12	2157696.03	25486128.86	

Customer purchase frequency

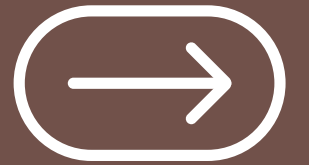


```
select customer_id, product_id,  
count(order_id) as Purchase_frequency  
from sales  
group by customer_id, product_id  
order by Purchase_frequency desc  
fetch first 15 rows only;
```



	CUSTOMER_ID	PRODUCT_ID	PURCHASE_FREQUENCY
1	C030798	P0090	5
2	C010143	P0161	4
3	C009169	P0057	4
4	C039287	P0027	4
5	C022340	P0173	4
6	C003616	P0093	4
7	C045454	P0015	4
8	C031163	P0010	4
9	C028885	P0050	4
10	C043263	P0022	4
11	C001386	P0161	4
12	C029292	P0142	4
13	C001006	P0132	4
14	C033959	P0169	4
15	C037167	P0142	4

Profit margin by product



```
select product_id, round(sum(profit) /  
sum(revenue)*100,2) as profit_margin  
from sales group by product_id order by  
profit_margin desc  
fetch first 15 rows only;
```



	PRODUCT_ID	PROFIT_MARGIN
1	P0109	40.25
2	P0063	40.25
3	P0108	40.24
4	P0067	40.23
5	P0139	40.18
6	P0133	40.18
7	P0037	40.17
8	P0081	40.17
9	P0060	40.16
10	P0061	40.16
11	P0165	40.16
12	P0053	40.15
13	P0003	40.15
14	P0079	40.15
15	P0166	40.15



Thank you!

